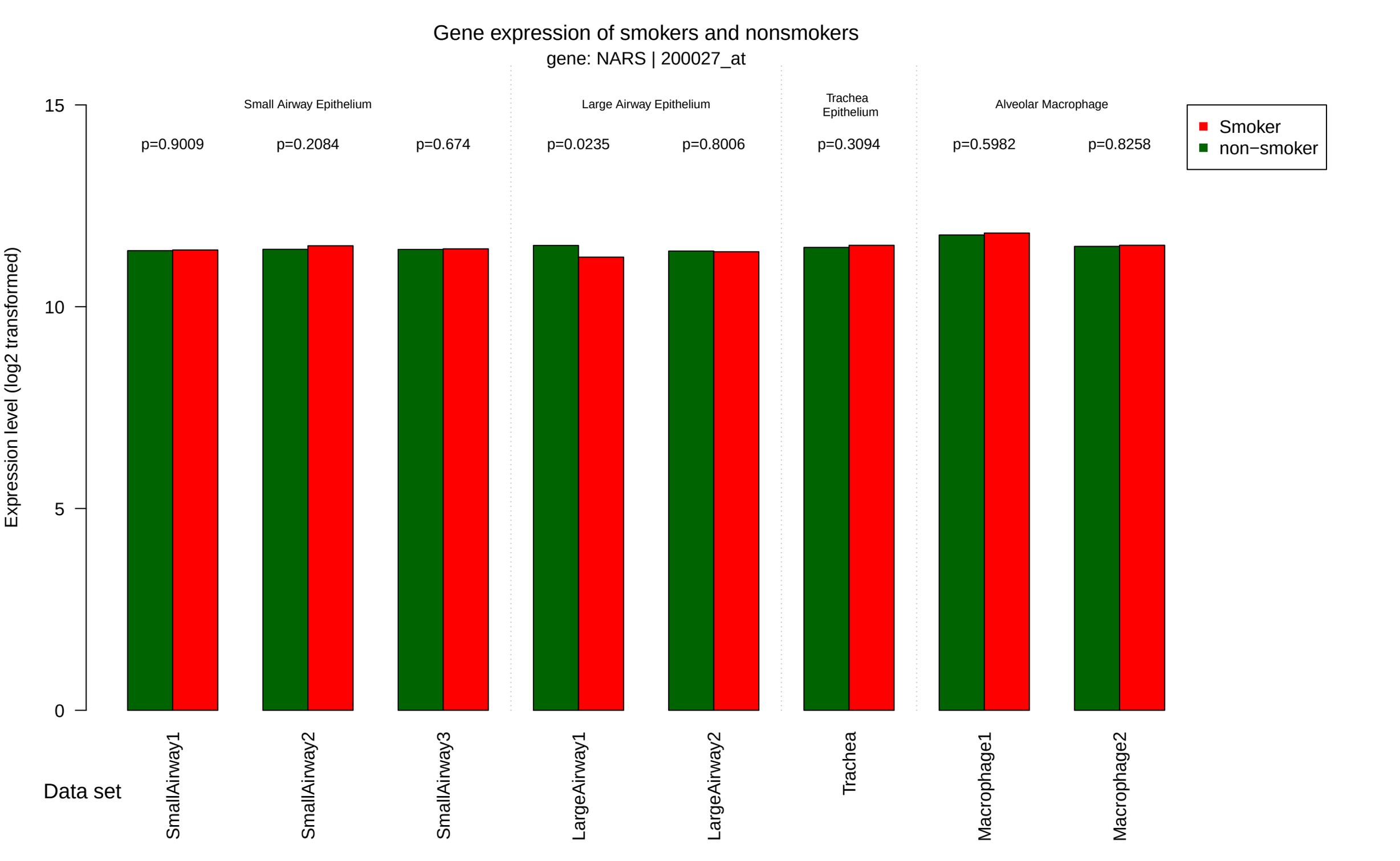
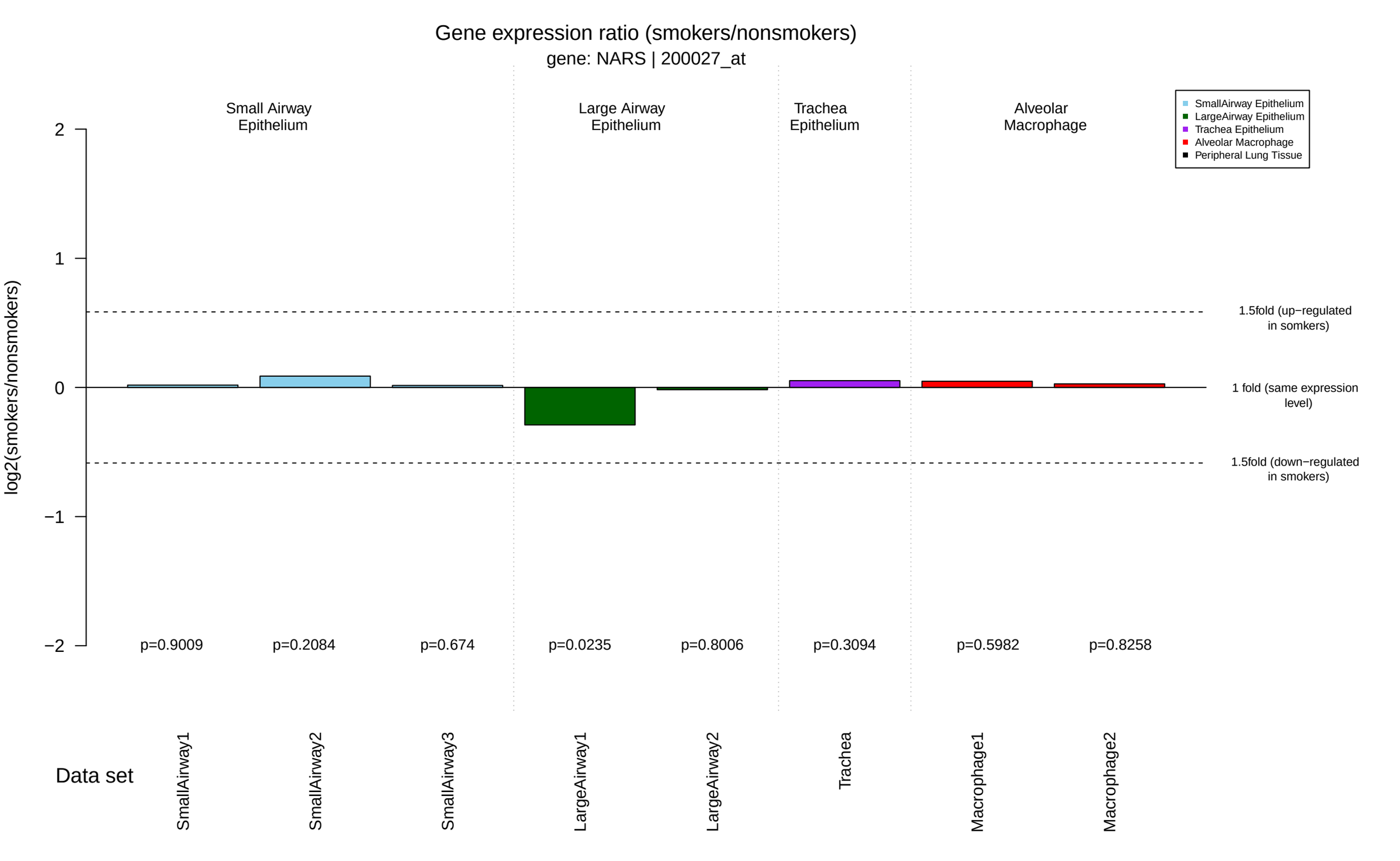
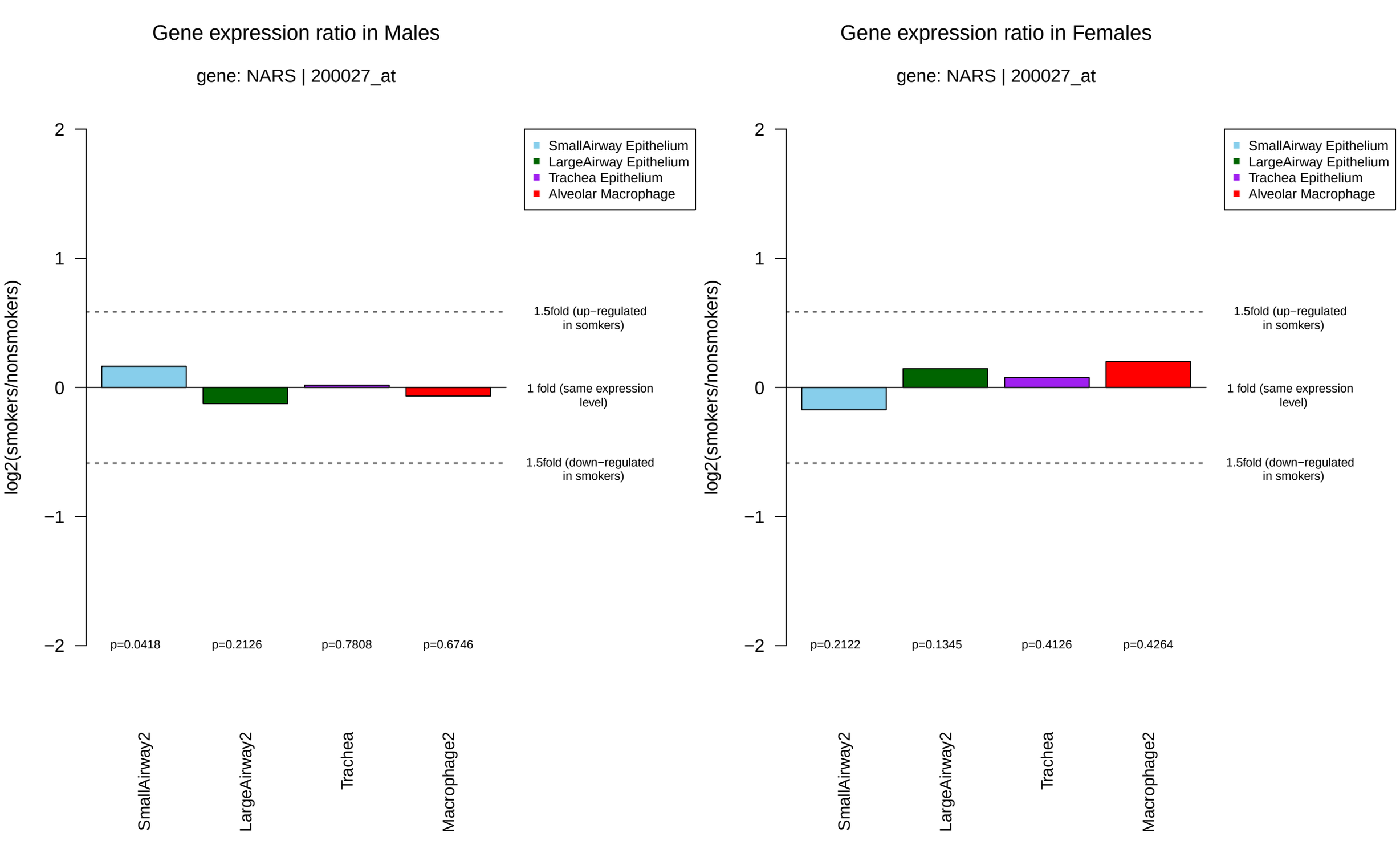
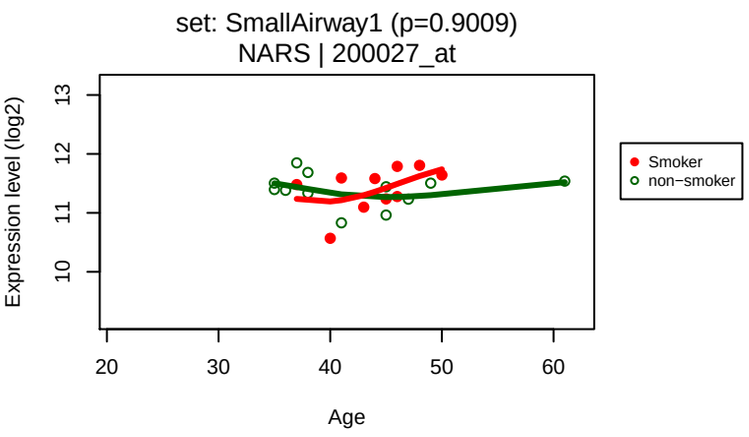
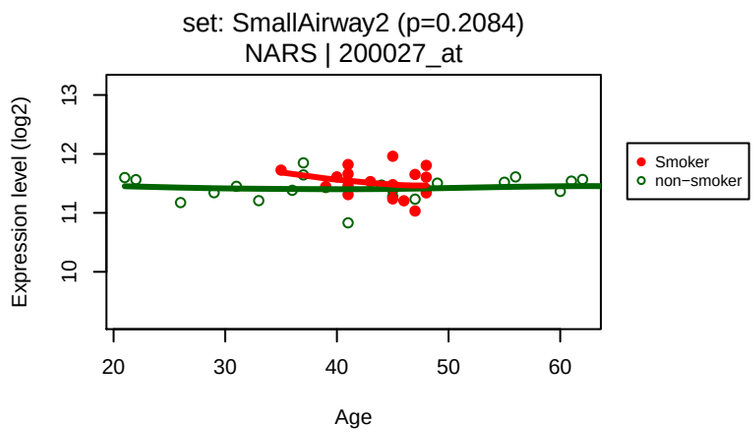
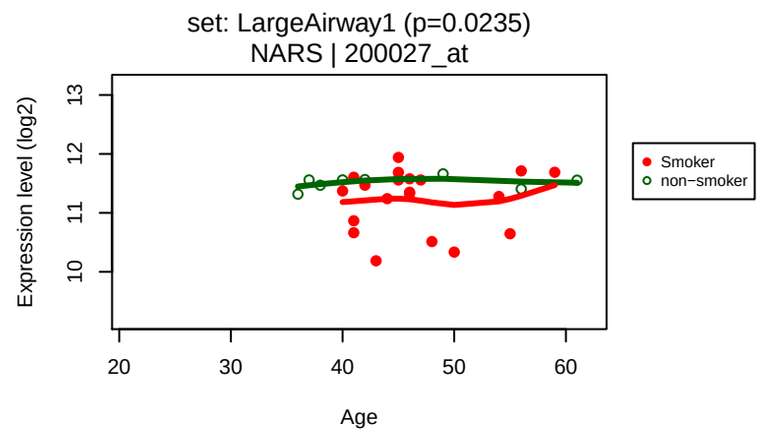

Gene expression of smokers and nonsmokers
gene: NARS | 200027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9009
p=0.2084
p=0.674
p=0.0235
p=0.8006
p=0.3094
p=0.5982
p=0.8258
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

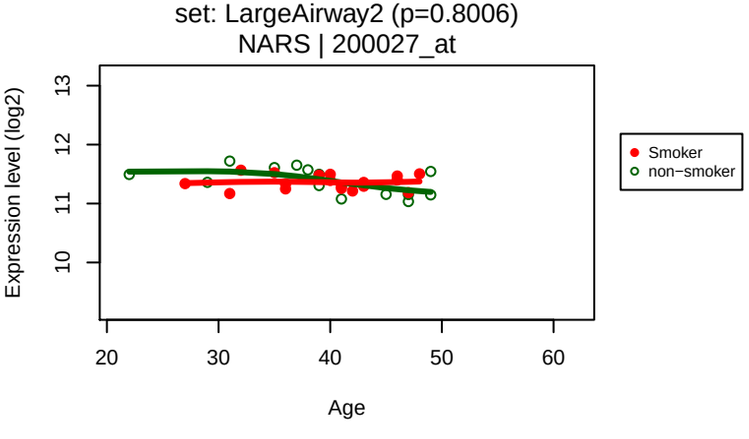

Gene expression ratio (smokers/nonsmokers)
gene: NARS | 200027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9009
p=0.2084
p=0.674
p=0.0235
p=0.8006
p=0.3094
p=0.5982
p=0.8258
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

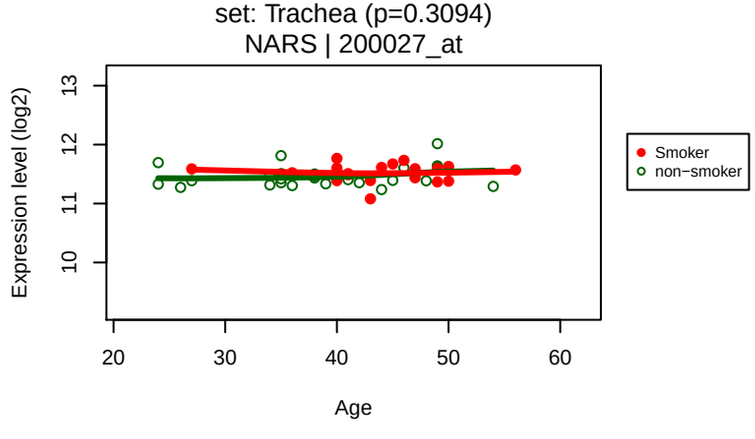

Gene expression ratio in Males
gene: NARS | 200027_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0418
p=0.2126
p=0.7808
p=0.6746
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NARS | 200027_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2122
p=0.1345
p=0.4126
p=0.4264
SmallAirway2
LargeAirway2
Trachea
Macrophage2

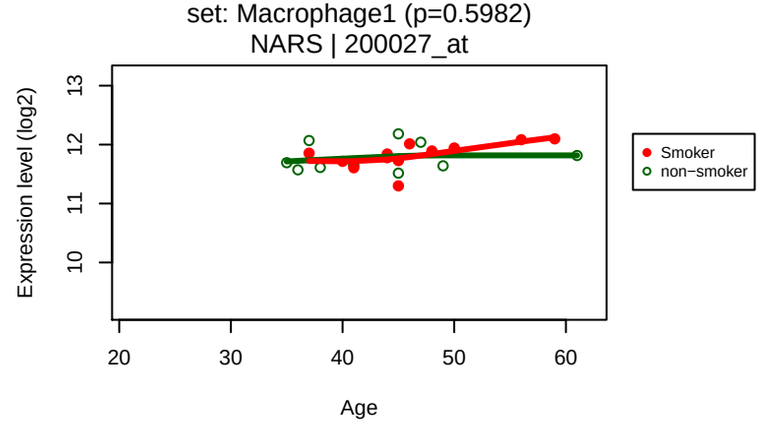

set: SmallAirway1 (p=0.9009)
NARS | 200027_at
Expression level (log2)
Age
Smoker
non−smoker

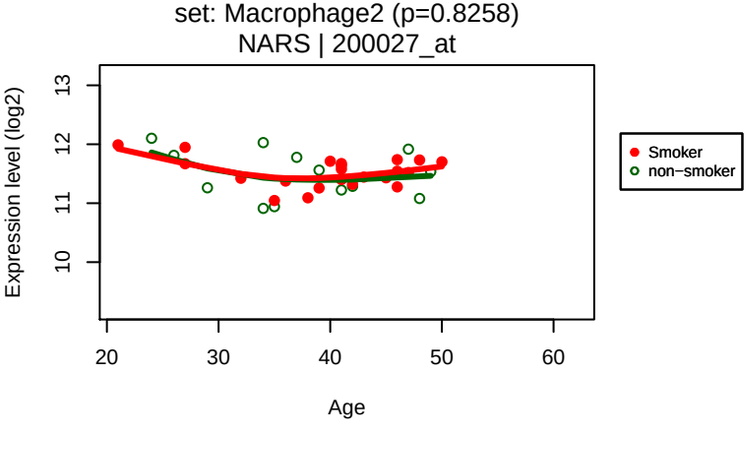

set: SmallAirway2 (p=0.2084)
NARS | 200027_at
Expression level (log2)
Age
Smoker
non−smoker

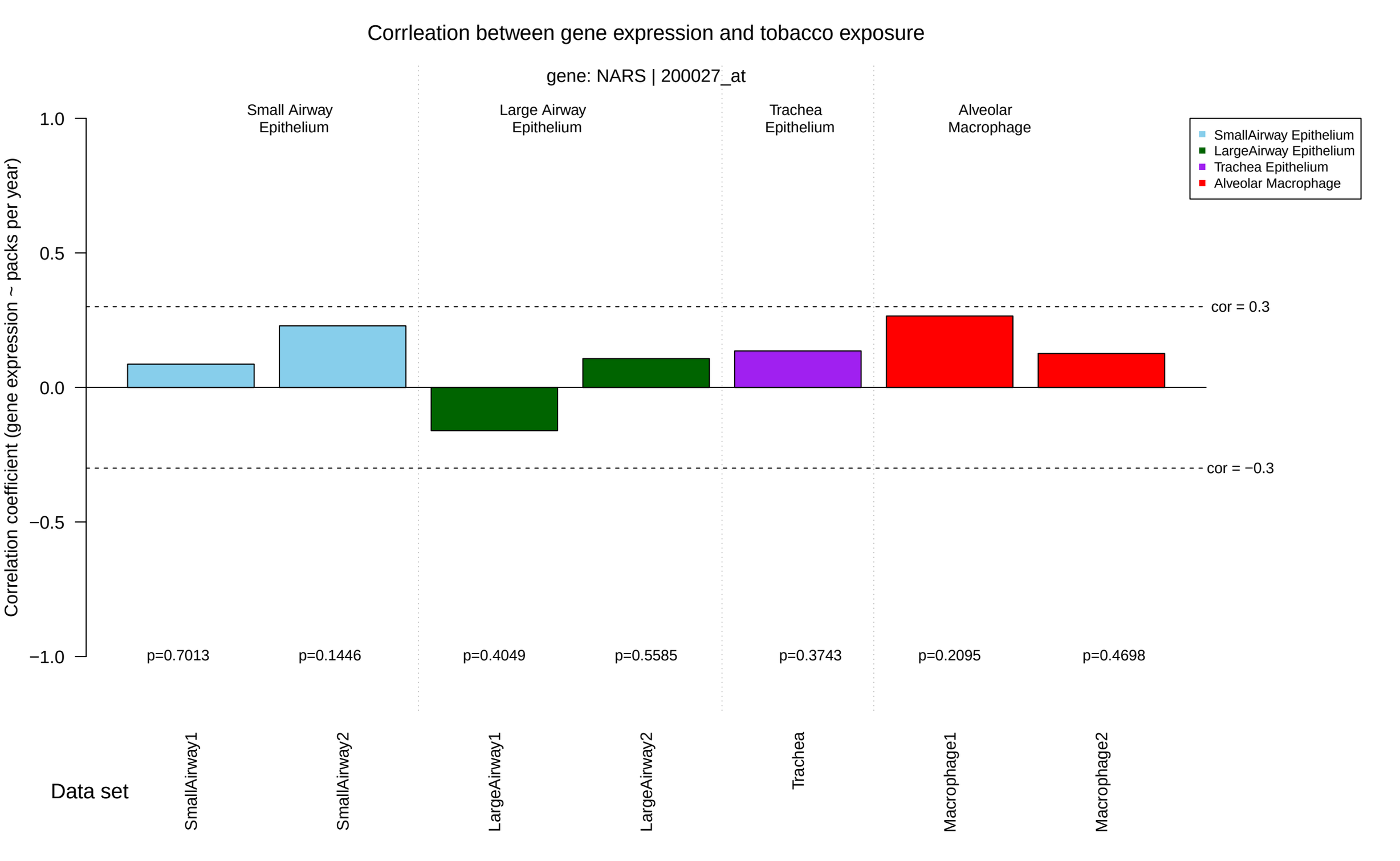

set: LargeAirway1 (p=0.0235)
NARS | 200027_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8006)
NARS | 200027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3094)
NARS | 200027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5982)
NARS | 200027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8258)
NARS | 200027_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NARS | 200027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7013
p=0.1446
p=0.4049
p=0.5585
p=0.3743
p=0.2095
p=0.4698
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2